NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Two Ways to Build an Interval

AP Statistics · Unit 3 · Topic 3.10 · B: 2026-12-11 · E: 2027-01-22

[STAR] TODAY'S PROBLEM

Town Oak has 2,200 households and Pine 1,800. Independent SRSs find 77 of 140 Oak households and 48 of 120 Pine households use curbside composting. Let $p_O - p_P$ mean Oak minus Pine.

Elena: "The two-sample z-interval is $(0.030, 0.270)$."

Devon: "Subtract the farthest endpoints of separate 95% intervals. That gives $(-0.020, 0.320)$, so zero is plausible."

Household samples

Town	Use	Other	n	N
Oak	77	63	140	2200
Pine	48	72	120	1800

FIRST TAKE — before the video (5 min)

The two reported ranges disagree about whether the towns could be equal. Which range would you trust, and why?

[BOOK] BUILD ONE INTERVAL FOR THE DIFFERENCE (keep on the page)

For two independent random samples, use a two-sample z-interval for $p_1 - p_2$: $(\hat{p}_1 - \hat{p}_2) \pm z^*\sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}$. For 95% confidence, $z^* = 1.96$. Check independent random samples or random assignment, each $n_i \le 0.10N_i$ when sampling without replacement, and at least 10 observed successes and failures in each group. Independent estimated variances add under one square root. Confidence belongs to the procedure used to construct the interval. An interval for a difference that includes 0 leaves equal proportions plausible; one entirely above 0 supports a positive difference.

Finish after the video:

DOK 1 **(a)** Define p_O and p_P and name the interval procedure for $p_O - p_P$.

DOK 2 **(b)** Check independent samples, both 10 percent conditions, and all four counts. Compute the estimate, SE, margin of error, and 95% interval.

DOK 3 **(c)** In 3–4 sentences, choose a method using the independent samples and the variance formula. Explain why subtracting separate 95% intervals does not establish a 95% interval for the difference. Use zero's location to explain how the method changes the conclusion.

Frame: The warranted method is ____________ because the data come from ____________________________, and the four counts are ___________________.

Frame: Devon's interval misleads a reader because ______________________________; as a result, zero appears __________ even though ___________________________.

EXIT — turn this in

One thing the video changed about my first take: ______________________________________